STAT 251 — Homework 4

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```r
# Global setup
set.seed(123)   # for reproducibility
digits <- 3     # inline values round to 3 digits
```

Question 1

> *The built-in `faithful` dataset records 272 eruptions of the Old Faithful geyser: the duration of each eruption and the waiting time until the next one (both in minutes).*

(a)

> *Report the mean and standard deviation of the waiting times.*

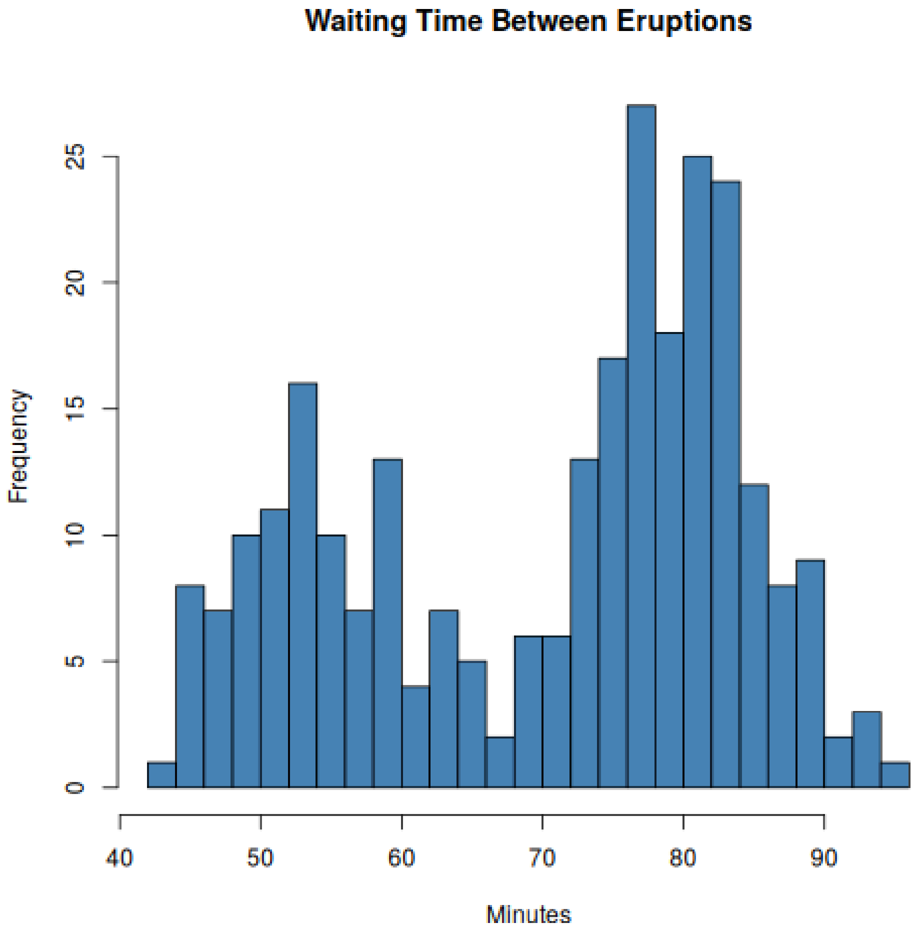
```r
wait <- faithful$waiting
n <- length(wait)
```

Across the 272 recorded eruptions, the mean waiting time is 70.897 minutes with a standard deviation of 13.595 minutes.

(b)

> *Plot a histogram of the waiting times. What do you notice?*

```r
hist(wait,
     breaks = 20,
     main = "Waiting Time Between Eruptions",
     xlab = "Minutes",
     col = "steelblue")
```



The distribution is clearly *bimodal*: eruptions cluster around short (about 55 min) and long (about 80 min) waits, so the mean alone is a poor summary of a "typical" wait.

Question 2

> *Suppose 65% of eruptions are "long" (waiting time above 68 minutes). If a visitor watches 8 independent eruptions, what is the probability that exactly 6 of them are long?*

Let X be the number of long eruptions. Then $X \sim \text{Binomial}(8, 0.65)$, so

$$P(X = 6) = \binom{8}{6}(0.65)^6(0.35)^2 \tag{1}$$

```
p6 <- dbinom(6, size = 8, prob = 0.65)
```

Evaluating Equation 1 gives $P(X = 6) = 0.259$.

Question 3

> *Construct a 95% confidence interval for the true mean waiting time.*

With $n = 272$ observations, the interval is

$$\overline{x} \pm t^*_{n-1} \frac{s}{\sqrt{n}} \tag{2}$$

```
ci <- t.test(wait)$conf.int   # results='hide': compute now, report inline
```

which evaluates to (69.274, 72.52) minutes. We are 95% confident the true mean waiting time lies in this interval — though given the bimodality from Question 1b, the *mean* wait is not a wait any visitor is likely to experience.

Question 4

> *Fit the simple linear regression of waiting time on eruption duration. State the model, report the fitted slope, and assess its significance.*

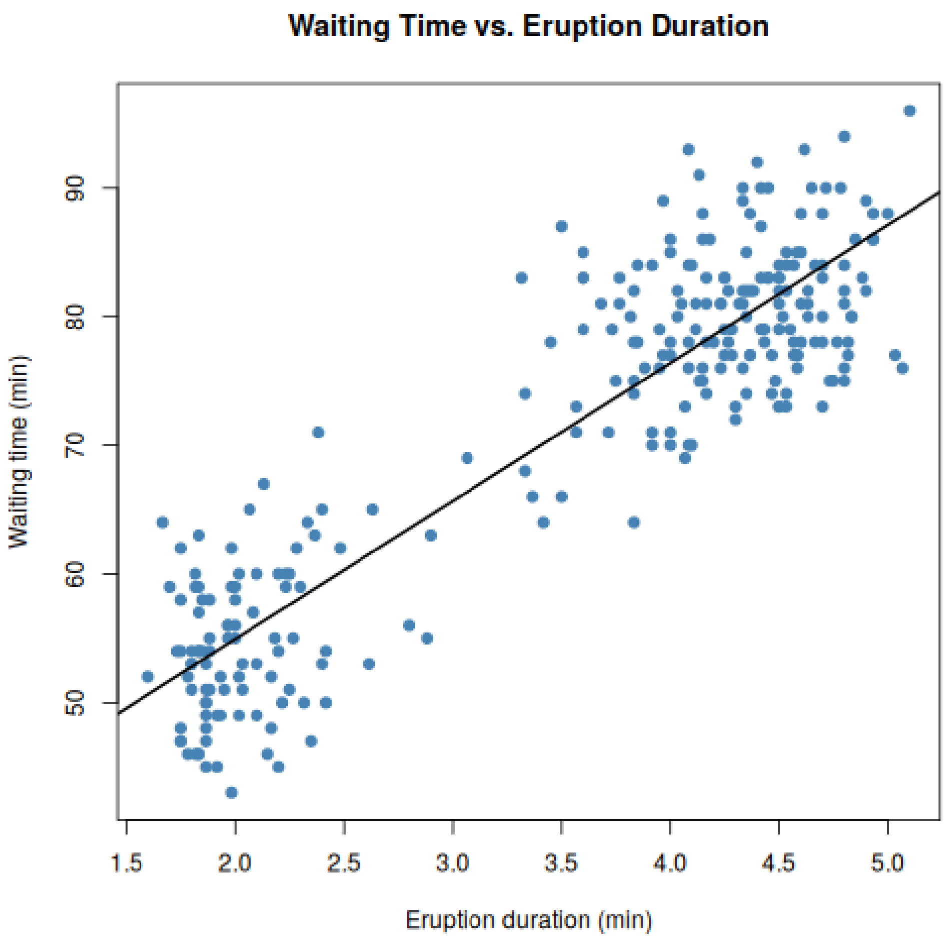
The model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2) \tag{3}$$

where Y_i is the waiting time and x_i the duration of eruption i.

```
fit <- lm(waiting ~ eruptions, data = faithful)
slope <- coef(fit)[2]
pval <- summary(fit)$coefficients[2, 4]

plot(faithful$eruptions, faithful$waiting,
     main = "Waiting Time vs. Eruption Duration",
     xlab = "Eruption duration (min)",
     ylab = "Waiting time (min)",
     pch = 19, col = "steelblue")
abline(fit, lwd = 2)
```



Each additional minute of eruption predicts a 10.73-minute longer wait. The slope is overwhelmingly significant ($p = 8.13 \times 10^{-100}$ — tiny inline values are rendered automatically in scientific notation), which matches the strong linear trend visible in the plot.

Appendix: tweave shorthand cheat sheet

All of these come from the `tweave` Typst package and work in any math block:

Write	**Get**	**Meaning**
`X iid "Normal"(mu, sigma^2)`	$X \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$	i.i.d. distribution statement
`bar(x)`	$\overline{x}$	sample mean (overline, not `|x`)
`choose(n, k)`	$\binom{n}{k}$	binomial coefficient (alias of `binom`)
`integral x f(x) dx`	$\int x f(x) \, \mathrm{d}x$	`dx`, `dy`, `dz`, `du`, `dv`, `dw`, `dtheta` render as differentials

For question numbering, prompt boxes, and other homework structure, tweave composes with anything: a homework template from Typst Universe, or a few `#let` definitions of your own (like `prompt` at the top of this file).